

DV300 Project Pitch

SymptomJournal

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Interactive Development 300

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SymptomJournal

"Log it calmly. Understand it clearly. Talk to your doctor with confidence."

Type: Mobile Application (iOS & Android - React Native)

Theme: Healthcare

Overview: SymptomJournal is an AI-powered mobile health companion that helps users log symptoms, identify patterns over time, and generate structured summaries for medical appointments - replacing the anxiety of Googling symptoms with calm, structured self-awareness.

Conceptualisation & Problem Statement

SymptomJournal is a mobile health companion designed to replace the anxiety of Googling symptoms with calm, structured self-awareness. The app guides users through a three-stage experience track, spot patterns, share with doctor mirroring the onboarding flow and continuing throughout the core app experience.

Problem Statement

In an age of instant information, people increasingly turn to the internet to understand their own health and it is making things worse. This behaviour, clinically documented as **cyberchondria**, is a modern extension of **hypochondria** the broader, long-standing condition of persistent health anxiety now amplified and triggered by compulsive online symptom searching that escalates worry rather than relieving it.

Research shows cyberchondria affects between 30.7% and 55.6% of studied populations, with roughly half of university students showing moderate levels and around a quarter scoring in the high range.

The problem extends beyond the home. Even when people do seek professional help, they struggle to communicate their symptoms accurately. Poor patient-doctor communication was identified as the sole cause of safety incidents in over 1 in 10 cases, and a contributing factor in 1 in 4 across a study of more than 67,000 patients. Patients forget symptom timelines, underestimate patterns, and walk out of short appointments without having communicated what they actually needed to say.

The gap is clear. Current options either make things worse symptom checkers and Google searches that suggest worst-case conditions or don't exist at all. No tool currently helps users track symptoms calmly over time, identify meaningful patterns, and arrive at a medical appointment with a clear, structured summary of their health history.

SymptomJournal addresses this directly. By combining a calm, supportive daily logging experience with AI-powered pattern analysis and structured doctor summary generation, it gives users agency over their health narrative without replacing the medical professional.

Target Audience

SymptomJournal is designed for anyone who has ever felt uncertain, anxious, or underprepared when it comes to their own health. Three distinct groups represent its core audience:

Young adults (18–30) experiencing health anxiety

Digitally native users who instinctively turn to the internet when something feels wrong and frequently spiral into worse anxiety as a result. They need a calmer, more structured alternative to Googling symptoms.

People managing chronic or recurring conditions

Users dealing with ongoing conditions migraines, IBS, fatigue, skin conditions who struggle to track patterns across weeks or months. The Symptom Timeline and AI Insights screens directly serve this group, surfacing connections like "headaches occur 82% more often after fewer than 6 hours of sleep."

Anyone who struggles to talk to their doctor

Whether due to anxiety, short appointment times, or simply forgetting symptoms in the moment many people leave medical appointments without having communicated what they needed to. The Doctor Summary screen generates a structured patient report covering entries logged, symptom types, frequency, average severity, and key patterns giving users something concrete to hand to their doctor.

What unites all three groups is a gap between what they experience and what they can effectively communicate and that is precisely what SymptomJournal is built to close.

SMART Objectives

Specific	Build a mobile health app that logs symptoms, detects AI patterns, and generates doctor summaries
Measurable	MVP complete with 6 core screens and 3 AI-powered features within 3 weeks of submission
Achievable	Built solo using free-tier tools React Native Expo, FastAPI, Supabase, Nvidia NIM
Relevant	Addresses documented cyberchondria problem affecting 30–55% of studied populations
Time-bound	Functional MVP deployed to Vercel and Railway within 21 days of pitch submission

Core Features

MVP - Must have

Feature	What it does
Daily symptoms log	User(s) log symptoms in plain language with optional severity rating (1 - 10) and time of day
AI patterns analysis	Analyses entries across time - Identifies clusters, triggers, frequency
Doctor summary generator	AI produces a structured, calm, jargon-free summary the user takes to their appointment
Symptom timeline view	Visual history of all logged entries - scrollable, filterable by symptom type
User Authentication	Secure login and personal data storage via Supabase Auth
Onboarding flow	First-time user setup - name, common conditions, notifications preferences

Nice to have

Feature	What is does
Trigger tracking	Log food, sleep quality, stress level alongside symptoms for richer AI analysis
PDF export	Download or share the doctor summary as a PDF
Daily reminder	Push notification prompting the user to log
Mood indicator	Quick emotional check-in alongside physical symptoms
Severity trend chart	Visual graph showing symptoms severity over time
Medication Info Assistant	AI-powered chatbot that allows users to look up information about medications prescribed by their doctor - strictly informational, never diagnostic

Future Considerations

Features for a later version:

Feature	What it does
React	Web application version for desktop
Multilingual support	SA's 11 official languages - Zulu, afrikaans, Xhosa priority
Telehealth share link	Shareable summary link for online doctor appointments
Wearable integration	Pull data from smartwatch health sensors
Family mode	Parent logging symptoms on behalf of their child

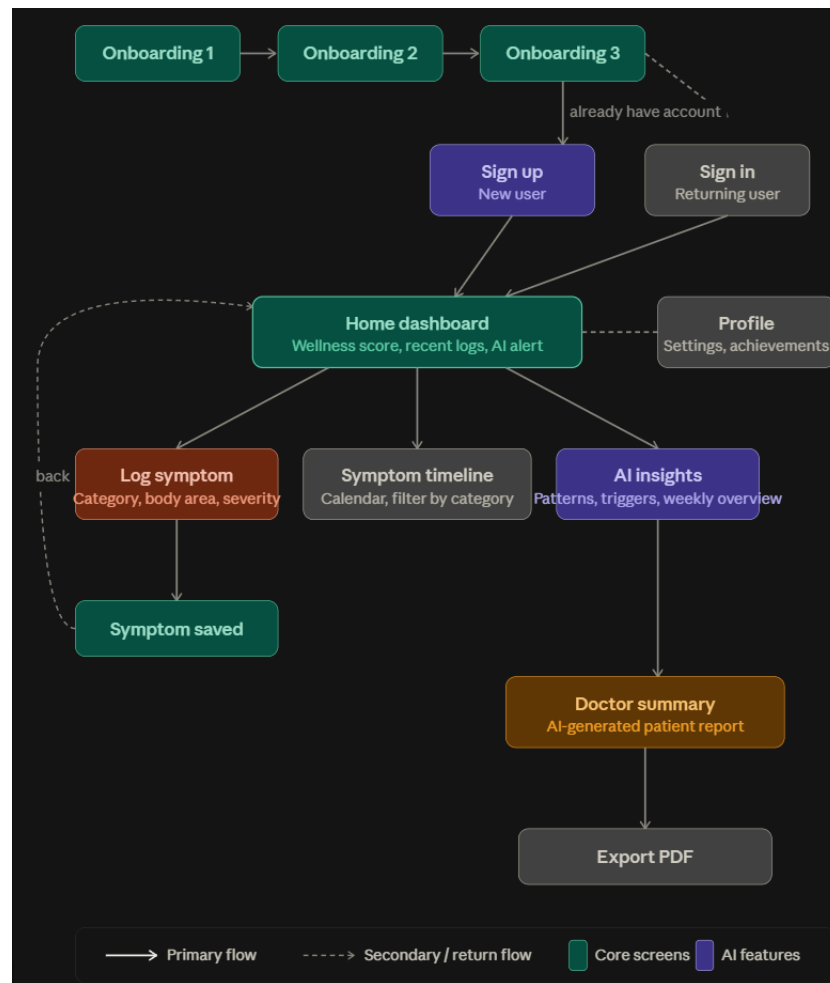
User Roles

Role	Access
Standard User	Full app access - log, view timeline, generate summaries
Guest	Onboarding and single session log only - no saved history

User Stories

- As a user with health anxiety. I want to log my symptoms calmly so I don't spiral into Googling them.
- As someone with recurring migraines, I want to see my trigger patterns so I can avoid them
- As a patient, I want a structured summary so my doctor understands my history in a short appointment
- As a new user, I want a gentle onboarding so the app doesn't feel clinical or intimidating.
- As a patient who has just received a new prescription, I want to look up my medication in the app so I can understand what it does and what side effects to expect.

User Flow



AI Technology

	Detail
AI Service	Nvidia NIM (build.nvidia.com) - primary. Google gemini API - fallback option
Models	Meta Llama 3.1 or Mistral via Nvidia NIM - strong natural language understanding, free tier
What AI does	Pattern analysis across symptoms entries, trigger identification, doctor summary generation, empathetic response tone.
Why AI is core	Without AI the app is a basic diary. Pattern detection across weeks of entries and coherent summary generation are impossible with a language model.
What happens without AI	Users get a log with no insight - no patterns identified, no doctor summary generated. The core value proposition disappears entirely.

Secondary AI Feature — Medication Information Assistant

SymptomJournal includes a strictly scoped medication information assistant. Users can enter the name of a medication prescribed by their doctor and receive clear, plain-language information about what it is, its common uses, and typical side effects. The AI is explicitly instructed never to suggest dosages, recommend alternatives, or comment on whether a prescription is appropriate. It always defers to the user's doctor for clinical decisions. This feature extends the app's core purpose — helping users communicate more effectively with their healthcare providers — without crossing into diagnostic territory.

Technologies & Stack

Layer	Technology	Why
Frontend	React Native (Expo)	Mobile-first, cross-platform iOS & Android, React knowledge transfers directly.
Backend	FastAPI (Python)	Industry standard for AI Backends, first-class Nvidia NIM SDK support
Database	Supabase	Free tier, Postgres, built-in auth, real-time capabilities
AI Engine	Nvidia NIM/ Gemini API	Free tier, production-quality models, OpenAI-compatible endpoints
Deployment	Vercel(Frontend) + Backend(Railway)	Free tier, zero cost
Version Control	GitHub	Industry standard, continuous progress tracking

Version Control

This project is version controlled using GitHub. The repository is available at:

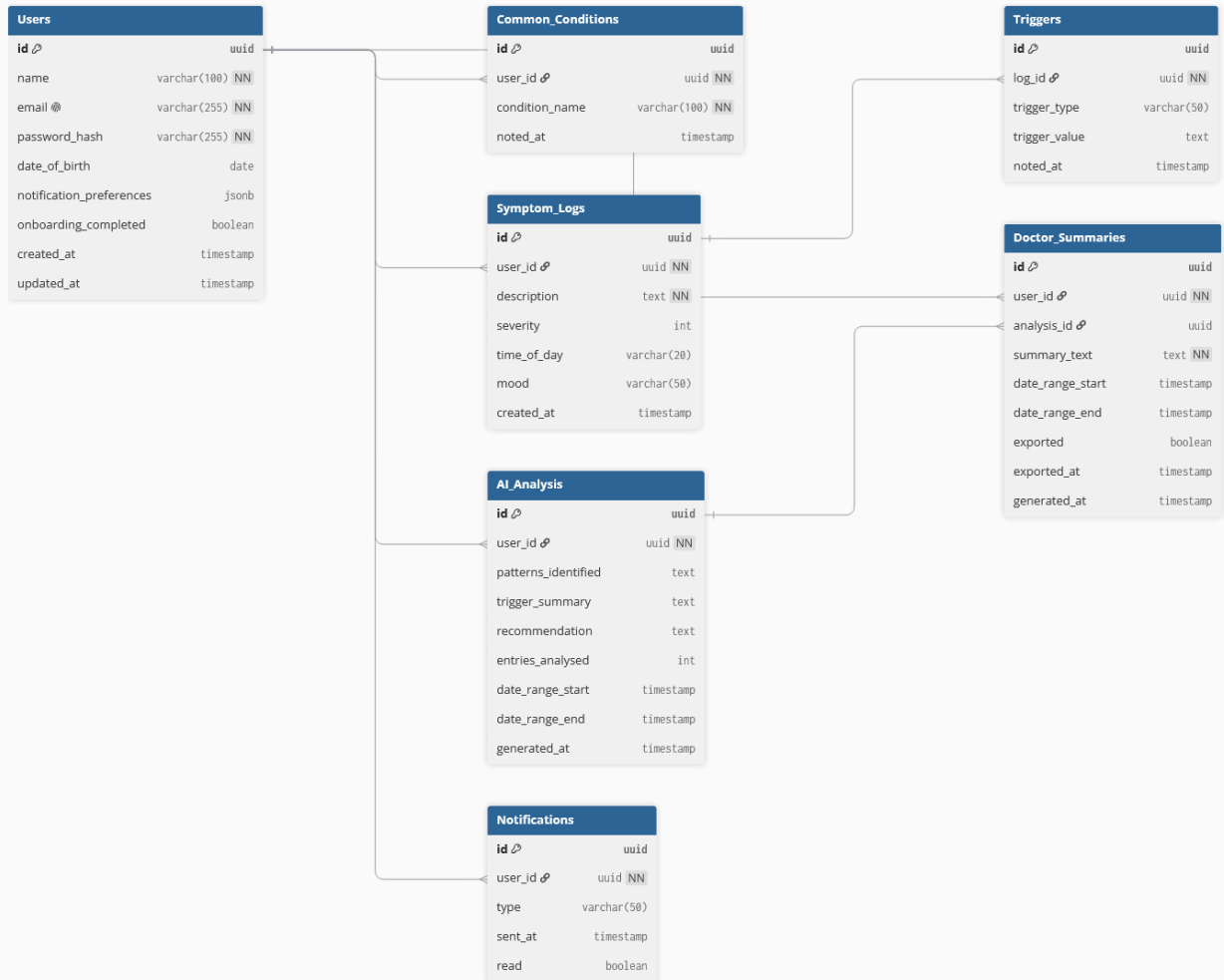
<https://github.com/FrancoisleRouxDev/SymptomJournal>

Commits follow Conventional Commits naming conventions — `feat:`, `fix:`, `chore:`, `docs:` — to maintain a clear and readable project history throughout development. Each development phase maps directly to a series of commits, making progress visible and traceable from day one.

Technical Challenges

Challenge	How I'll Approach it
AI returns inconsistent responses	Structured prompting with a strict output format instruction + response validation layer in FastAPI
Keeping AI tone calm and non-alarmist	System prompt engineering - explicitly instructing the model never to suggest conditions or medications
Pattern analysis needing enough data	Minimum entry threshold before AI analysis triggers - prompt user to log in more if insufficient data
Mobile performance with API calls	Loading states, optimistic UI updates, and caching recent AI results locally
Data privacy and sensitivity	All health data stored in Supabase with row-level security users only ever access their own data
Python being a new language	Learning FastAPI fundamentals during holiday period before semester starts

Database Structure - ERD Overview



Brand & Visual Identity

Wireframes & UI/UX Considerations

Design Inspiration

The visual direction draws from calm wellness apps and health journaling tools prioritising warmth and readability over clinical sterility. References included mental health app UI kits, warm sage and terracotta colour combinations, and minimal health tracking interfaces.



Typography

Aa

DM Serif Display
Headings & display

Aa

Nunito
Body & UI

TYPE SCALE

Display
32px / 400

Good morning, Sarah

Heading 1
26px / 400

Symptom Timeline

Heading 2
22px / 400

Weekly Insights

Label
11px / 700

CATEGORY · UPPERCASE

Body Bold
14px / 700

Tension Headache

Body
14px / 500

Log symptoms in seconds and build a clear picture of your health over time.

Caption
12px / 500

Today, 2:30 PM · Head

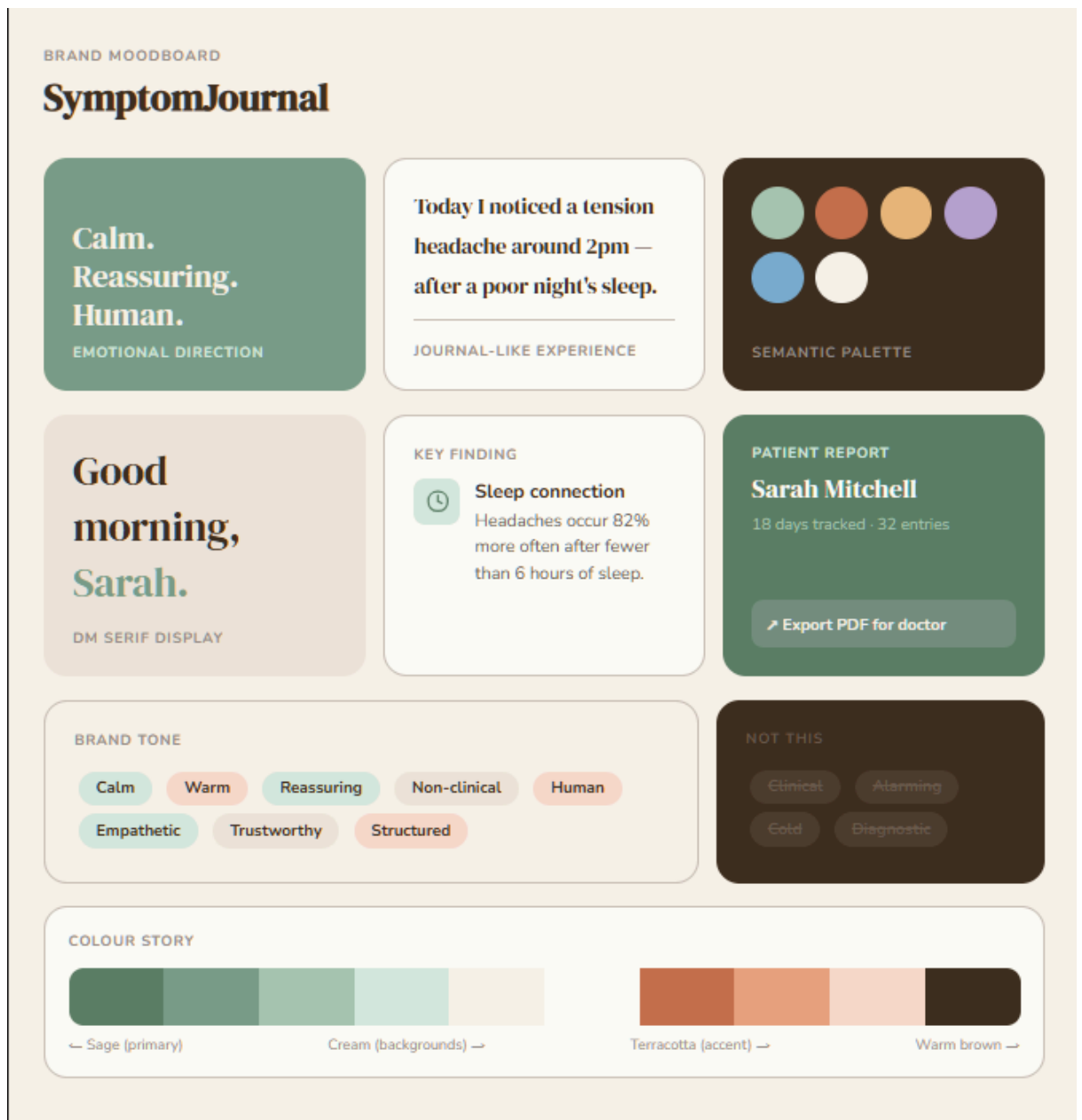
Micro
10px / 700

Mild · Moderate · Severe

Button
16px / 700

Save Symptom

Moodboard



Logos



UI Components

- Category tiles with emoji icons on Log a Symptom screen quick selection without reading
- Severity slider with colour gradient (mild green → severe red) intuitive without numbers alone
- Card-based AI insights patterns surface as readable findings, not raw data
- Dark patient report header on Doctor Summary signals document mode, clinical credibility
- Terracotta CTA buttons warm but attention-grabbing against cream backgrounds

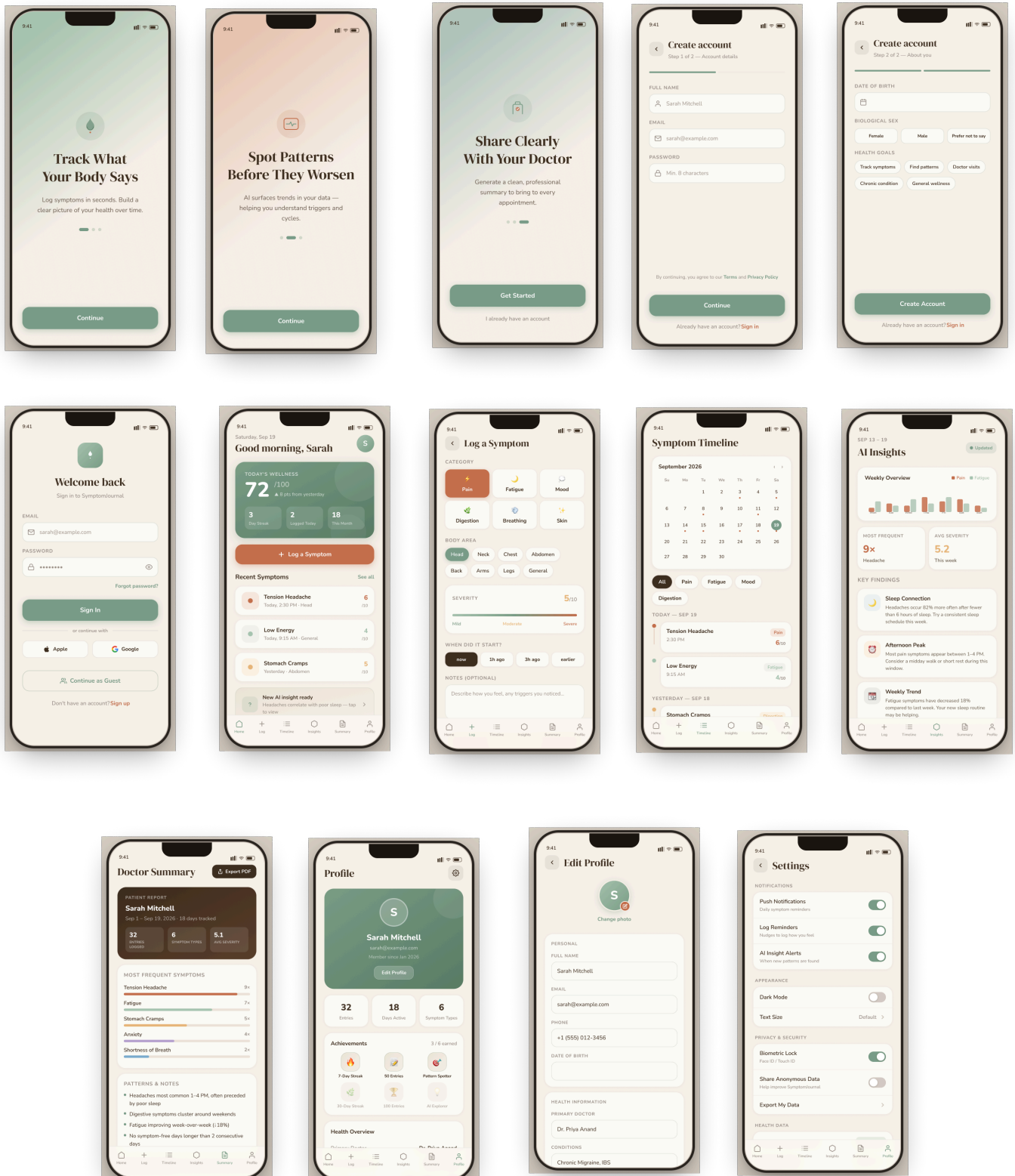
Accessibility

- High contrast text on all backgrounds
- Large touch targets on all interactive elements
- Colour never used as the sole indicator icons and labels accompany all colour coding
- Text size adjustable in Settings
- Biometric lock available for sensitive health data privacy

Usability

- Six-tab bottom navigation consistent across all main screens
- Three-screen onboarding introduces all three core value propositions before sign-up
- "When did it start?" time selector on log screen reduces cognitive load
- Empty state handling first-time users guided to log their first symptom immediately after onboarding
- AI insights framed as findings, not chatbot responses avoids recreating the cyberchondria dynamic inside the app

Wireframes



Project Roadmap

Phase	What happens	Timeline
Phase 1 - Setup	Repo scaffolded, Python fundamentals, expo project initialised, Supabase project created, environment configured	Days 1 - 3
Phase 2 - Foundation	Auth flow, onboarding screens, basic symptom logging UI, Supabase tables connected	Days 4–7
Phase 3 AI Integration	FastAPI backend, Nvidia NIM connection, prompt engineering, pattern analysis endpoint, doctor summary endpoint	Days 8–12
Phase 4 Core Features	Timeline view, severity ratings, trigger logging, doctor summary screen, AI results display	Days 13–16
Phase 5 Polish & Testing	Brand identity applied, UI consistency, AI response validation, edge cases, bug fixes	Days 17–19
Phase 6 Deployment	Build production APK using Expo EAS Build, upload to GitHub releases. Share Expo Go link for testing. Final demo preparation.	Days 20–21

Conclusion

Health anxiety is not a niche problem it affects millions of people globally, and the internet has made it significantly worse. Cyberchondria, the compulsive cycle of online symptom searching that escalates rather than relieves anxiety, now affects between 30% and 55% of studied populations. At the same time, poor patient-doctor communication contributes to safety incidents in 1 in 4 medical cases not because patients don't care, but because they lack the tools to track and communicate their health history effectively.

SymptomJournal addresses both problems directly. By replacing the Google search spiral with a calm, structured daily logging experience and using AI to find the patterns users can't see themselves it gives people agency over their own health narrative. The AI-generated doctor summary closes the communication gap, turning a short, stressful appointment into a productive conversation.

The technology stack is entirely free and open source, the scope is realistic for a single semester, and the problem is both well-researched and personally relevant to the target audience. AI is not a feature added to satisfy a brief requirement it is the core of what makes this application valuable. Remove it and the app does nothing meaningful.

SymptomJournal is a project worth building because the problem it solves is real, the users who need it are clearly defined, and the solution is something that could genuinely improve how people manage and communicate their health.

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